Reinforcement Learning With Reward Machines in Stochastic Games

Reinforcement Learning Course - Final Project — 06-2026

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Problem Setting

- **Problem:** How to tackle complex tasks in **multi-agent RL** where reward functions are **non-Markovian**?
- **Proposed Approach:** Usage of **Reward Machines** to incorporate the high-level knowledge of the task. By RM state incorporation, form an augmented product stochastic game, **converting non-Markovian rewards** into **standard Markovian ones** [1].
- **Goal:** Learn and converge to the optimal best-response strategy at a **Nash equilibrium** for each agent in a **general-sum game setting**.

Basic Definitions

Stochastic Games with Non-Markovian Rewards

Definition — Two-Player General-Sum Stochastic Game (SG)

Modeled as the tuple $\mathcal{G} = (S, s_I, A_\mathfrak{e}, A_\mathfrak{a}, p, R_\mathfrak{e}, R_\mathfrak{a}, \gamma)$, where:

- **Finite State Space (S):** consisting of states $s = [s_\mathfrak{e}; s_\mathfrak{a}]$, with $s_\mathfrak{e}, s_\mathfrak{a}$ substates for *Ego* and *Adv* agents respectively. $s_I \in S$ is the initial state.
- **Action Spaces:** Each agent has its own finite set of actions $A_\mathfrak{e}$ and $A_\mathfrak{a}$.
- **Probabilistic Transition Function**: $p : S \times A_\mathfrak{e} \times A_\mathfrak{a} \times S \to [0, 1]$
- **Non*-Markovian Reward functions**: $R_{\mathfrak{e} \text{ or } \mathfrak{a}} : (S \times A_\mathfrak{e} \times A_\mathfrak{a})^+ \times S \to \mathbb{R}$ map from **complete trajectory histories** to real value domains, specifying the n-m rewards for both agents.

(*) Usual definition assumes system Markovianity.

Trajectories and Rewards

- **Trajectory:** Sequence of states and actions $t_{0:k+1} = s_0 a_{\mathfrak{e},0} a_{\mathfrak{a},0} s_1 ... s_k a_{\mathfrak{e},k} a_{\mathfrak{a},k} s_{k+1}$ with $s_0 = s_I$.
- **Reward Sequence:** Given a **trajectory**, each agent $(i \in \{\mathfrak{e}, \mathfrak{a}\})$ receives reward sequence $r_{i,1} ... r_{i,k}$, where $r_{i,k} = R(s_0, a_{\mathfrak{e},0}, a_{\mathfrak{a},0}, ..., s_k)$.
- **Shared Observation:** Both agents **observe the same** trajectory.
- **Discounted Cumulative Reward:** Given a **trajectory** $t_{0:k+1}$, agents achieves a DCR reward $\sum_{t=0}^{k} \gamma^t r_{\mathfrak{i},t}$
- **Finite episodes assumption**: The game is episodic with a finite horizon, ensuring that the cumulative reward is well-defined.

Strategies and Objectives

- **Markovian Strategy:** Agents are equipped with a $\pi_i : S \times A_i \to [0,1] \quad i \in \{\mathfrak{e}, \mathfrak{a}\}$ strategy, defining action given state selection probability.
- **Objective:** Maximize the **expected** discounted cumulative reward given policy of both agents, from both point of view.

Definition — SG Expected Discounted Cumulative Reward

Given agent $i \in \{\mathfrak{e}, \mathfrak{a}\}$, the expected DCR under policies $\pi_\mathfrak{e}$ and $\pi_\mathfrak{a}$ is defined as:

$$\tilde{v}_i(s, \pi_\mathfrak{e}, \pi_\mathfrak{a}) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}(r_{i,t} | \pi_\mathfrak{e}, \pi_\mathfrak{a}, s_0 = s)$$

Nash Equilibrium

Each agent's strategy is a **best-response to the other agent's strategy**, if each agent **cannot improve its own reward** by changing its strategy, given a **fixed opponent's policy**.

Definition — Nash Equilibrium in Stochastic Games

For a stochastic game $\mathcal{G} = (S, s_I, A_\mathfrak{e}, A_\mathfrak{a}, R_\mathfrak{e}, R_\mathfrak{a}, \gamma, p)$, the strategies of the ego agent, $\pi^*_\mathfrak{e}$, and the adversarial agent, $\pi^*_\mathfrak{a}$, are at a **Nash equilibrium** if:

$$\tilde{v}_\mathfrak{e}(s, \pi^*_\mathfrak{e}, \pi^*_\mathfrak{a}) \geq \tilde{v}_\mathfrak{e}(s, \pi_\mathfrak{e}, \pi^*_\mathfrak{a})$$
$$\tilde{v}_\mathfrak{a}(s, \pi^*_\mathfrak{e}, \pi^*_\mathfrak{a}) \geq \tilde{v}_\mathfrak{a}(s, \pi^*_\mathfrak{e}, \pi_\mathfrak{a})$$

hold for any $\pi_\mathfrak{e}$ and $\pi_\mathfrak{a}$.

Objectives (cont'd)

The **actual objective** of each agent is to find the policy π_i^*, maximising the **expected discounted cumulative reward** considering the other agent's action once it reaches the game's Nash equilibrium

Agent Objectives at Nash Equilibrium

$$\pi_{\mathfrak{e}}^* = \arg\max_{\pi_{\mathfrak{e}}} \tilde{v}_i(s, \pi_{\mathfrak{e}}, \pi_{\mathfrak{a}}^*)$$

$$\pi_{\mathfrak{a}}^* = \arg\max_{\pi_{\mathfrak{a}}} \tilde{v}_i(s, \pi_{\mathfrak{e}}^*, \pi_{\mathfrak{a}})$$

Reward Machines

Definition — Reward Machine (RM)

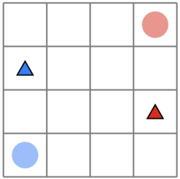
Finite state automaton designed to encode non-Markovian rewards with state transitions and output functions: $\mathcal{A} = (V, v_I, 2^{\mathcal{P}}, \mathbb{R}, \delta, \sigma)$

- **Input Alphabet ($2^{\mathcal{P}}$):** Interprets propositional logic variables $\mathcal{P}$ generated from lower-level environment updates.
- **Set of RM states (V):** Finite collection of internal tracking states, with v_I as the initial state.
- **Transition function:** Deterministic state progression $\delta : V \times 2^{\mathcal{P}} \rightarrow V$.
- **Output function:** Discrete scalars along active trajectory $\sigma : V \times 2^{\mathcal{P}} \rightarrow \mathbb{R}$.

Reward Machines example visualization

We define Reward Machines as a formalism to express **task specification**, decomposed into **several stages**.

Example — variant of the PAC-MAN game



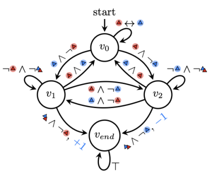


Figure 1.,2. *Variant of the PAC-MAN game is a symmetric zero-sum game where two agents try to capture each other. [1]*

Connecting SG and RMs

A **link between RM and SG** is required

Labelling Function

$$L : S \times A_{\mathfrak{e}} \times A_{\mathfrak{a}} \times S \to 2^{\mathcal{P}}$$

$l_t = L(s_t, a_{\mathfrak{a},t}, a_{\mathfrak{e},t}, s_{t+1})$ is the set of **true propositions** in $\mathcal{P}$ at step t.

RM can encode non-Markovian RF of a stochastic game

Given a SG, the RM $\mathcal{A}_i$ **encodes** the non-markovian reward function R_i of the game, meaning that **for each trajectory** $s_0 a_{\mathfrak{e},0} a_{\mathfrak{a},0} s_1 \ldots s_k a_{\mathfrak{e},k} a_{\mathfrak{a},k} s_{k+1}$ (i.e. $l_0 l_1 \ldots, l_k$) it outputs the **reward sequence** $r_{i,0} r_{i,1} \ldots, r_{i,k}$.

I.e. the RM maps **any sequence of events to a sequence of rewards**.

Q-learning with RM for SG

Problem formulation

The paper focuses on **two-agent general-sum stochastic games** (SG) with NM reward functions, where each agent is given a (different) task to complete.

Task completition **depends on the other's agent behavior**.

- Each agent as a separate reward machine.
- Agents operate in a shared and adversarial environment.

Then, the SG with non-markovian reward functions is formalized as **a product stochastic game** to obtain **Markovian rewards** using **reward machines**.

Markovian formulation: SGRM

Definition — Stochastic Game with Reward Machines (SGRM)

Given an SG $\mathcal{G}$ and RMs $\mathcal{A}_{\mathfrak{e}}, \mathcal{A}_{\mathfrak{a}}$, a SGRM is defined as a product stochastic game:

$$\mathcal{H} = (S', s'_I, A_{\mathfrak{e}}, A_{\mathfrak{a}}, p', R'_{\mathfrak{e}}, R'_{\mathfrak{a}}, \gamma)$$

- **Augmented State Space:** $S' = S \times V_{\mathfrak{e}} \times V_{\mathfrak{a}}$. An **augmented state** is $s' = (s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$.
- **Transition Probabilities:** $p'(s', a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s}') = p(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s})$ if $\bar{v}_i = \delta_i(v_i, L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s}))$ for both $i \in \{\mathfrak{e}, \mathfrak{a}\}$, and 0 otherwise.
- **Markovian Reward Functions:** $R'_i(s', a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s}') = \sigma_i(v_i, L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, \bar{s})), \quad \forall i$

The **SGRM** is proved to have a Markovian **RF** to express the original **non-markovian RF** of the **SG**.

SGRM Steps

The agent consider S' to select actions: at **each time step simultaneous action selection and execution** occurs; this causes the **game state transition** from s to s'.

The SGRM labelling function L returns the **high-level event associated**, l_t. Using current label, the RM performs **RM state transition**, by moving from $(v_{\mathfrak{e}}, v_{\mathfrak{a}})$ to $(v'_{\mathfrak{e}}, v'_{\mathfrak{a}})$, using each agent's specific RM, i.e. $v'_{\mathfrak{i}} = \delta(v_{\mathfrak{i}}, L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, s')$

The **transition of both game states and RM states** leads to the step **reward** for the agent $r_{i,t} = \sigma(v_{\mathfrak{i}}, L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, s')$

Nash Equilibrium and Q-functions in SGRM

Q-functions are used at Nash equilibrium.

QSG-RM

Q-function at a Nash equilibrium is the EDCR obtained by agent i when **both agents** follw a **joint** Nash Equilibrium strategy from the next period on.

So, it depends on **both agents actions**, and it's defined over the augmented state space.

Definition — Q-Function at a Nash Equilibrium in SGRM

Given agent i, given augmented state space $S \times V_{\mathfrak{e}} \times V_{\mathfrak{a}}$, we define

$$q_i^*(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) = r_i + \gamma \sum_{s', v'} p(s', v' | s, v, a_{\mathfrak{e}}, a_{\mathfrak{a}}) \tilde{v}_i(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}}, \pi^*_{\mathfrak{e}}, \pi^*_{\mathfrak{a}})$$

Nash Equilibrium and Q-functions in SGRM (cont'd)

Using the q-function in the augmented space, best responce policies are then originally extracted using the **Lemke-Howson method**.

By definition an agent strategy is optimal if the other agent behavior is considered: this requires both agents to **maintain two Q-functions**.

Agent learned (two) Q-functions

$q_{i,j}$ represents the Q-function for agent i as learned by agent j. It represents the **estimation that agent j** does of agent **i's expected discounted cumulative reward**, assuming a game-theoretic baseline where each plays **accordingly to Nash Equilibrium** (i.e. choosing the best response to the other player -estimated-strategy.)

As a consequence, **each agent can estimate** (its own and) **other player's strategy** $\pi_{i,j}$, using the q-function estimation of its EDCR.

The Stage Game Matrix

Definition: Stage Game

A two-player stage game is defined as $(r_{\mathfrak{e}}, r_{\mathfrak{a}})$, where r_i is agent i's reward function R_i, over the space of joint actions:

$$r_i = \{R_i(a_{\mathfrak{a}}, a_{\mathfrak{e}}) | a_{\mathfrak{a}} \in A_{\mathfrak{a}}, a_{\mathfrak{e}} \in A_{\mathfrak{e}}\}$$

- In QRM-SG, a stage game is formulated at each time step based on current Q-functions in augmented state space.
- The **Lemke-Howson method** is used to derive best-response strategies at a Nash equilibrium.

QSG-RM Algorithm Overview

QRM-SG Algorithm Key aspects

Actor Mechanics (Action Selection)

- ϵ-greedy policy is adopted to balance the **exploration** and **exploitation**.

$$a_i \leftarrow \begin{cases} \text{argmax}_{a \in A_i} \pi_{ii}(a \mid s, v_{\mathfrak{e}}, v_{\mathfrak{a}}) & \text{with probability } 1 - \epsilon \\ \text{random} & \text{with probability } \epsilon \end{cases}$$

Critic Updates (Learning Loop)

Values are updated according to standard stochastic approximation:

$$q_{ji} \leftarrow (1 - \alpha_t) q_{ji} + \alpha_t (r_{i,t} + \gamma \overline{q}_{ji}(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}}))$$

where

$$\bar{q}_{ji} \leftarrow \pi_{\mathfrak{e},\mathfrak{i}}(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}}) \pi_{\mathfrak{a},\mathfrak{i}}(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}}) q_{ji}(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}}) \in \mathbb{R}$$

QRM-SG Phase 1 — Initialization

Process Breakdown (Lines 1–4)

Hyperparameter: episode length *eplength*, γ, ϵ
Input: Reward machines $\mathcal{A}_{\mathfrak{e}}, \mathcal{A}_{\mathfrak{a}}$

$s \leftarrow \mathit{InitialState}()$; $v_{\mathfrak{e}} \leftarrow v_{I,\mathfrak{e}}$; $v_{\mathfrak{a}} \leftarrow v_{I,\mathfrak{a}}$
$q_{\mathfrak{e}i}(\cdot), q_{\mathfrak{a}i}(\cdot) \leftarrow \mathit{InitialQfunctions}()$

Analysis:

- Each agent initializes **two** Q-tables to store values for self and opponent, allowing local equilibrium computation.
- A SGRM (augmented) states are defined in the product space $S \times V_{\mathfrak{e}} \times V_{\mathfrak{a}}$; each tuple element s, $v_{\mathfrak{e}}$ and $v_{\mathfrak{a}}$ is **individually initialized**.

QRM-SG Phase 2 — Nash-Action Selection

Process Breakdown (Lines 7–13)

for each episode, for each time step t:

$\bar{\epsilon} \leftarrow \mathit{GenerateRandomValue}(0, 1)$

if $\bar{\epsilon} < \epsilon$ **then**

  $a_i \leftarrow \mathit{ChooseActionRandomly}()$ # [Up, Down, Left, Right]

else

  for $i \in \{\mathfrak{e}, \mathfrak{a}\}$ **do**

    $\pi_{\mathfrak{e}i}, \pi_{\mathfrak{a}i} \leftarrow \mathit{CalculateNashEquilibrium}(q_{\mathfrak{e}i}, q_{\mathfrak{a}i})$

    $a_i \leftarrow \text{argmax } \pi_{ii}(a|s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$

  end for

end if

At **each step**, agents solve a bimatrix stage game using the **Lemke-Howson method** to find mixed strategy equilibria.

QRM-SG Phase 3 — Reward Machine Dynamics

Previously selected step action gets executed:

Process Breakdown (Lines 14–18)

$s' \leftarrow \mathit{ExecuteAction}(s, a_{\mathfrak{e}}, a_{\mathfrak{a}})$ # SG state transition, i.e. new coords
$l_t \leftarrow L(s, a_{\mathfrak{e}}, a_{\mathfrak{a}}, s')$
for $i \in \{\mathfrak{e}, \mathfrak{a}\}$ **do**
  $v'_i \leftarrow \delta_i(v_i, l_t); r_{i,t} \leftarrow \sigma_i(v_i, l_t)$
end for

- L bridges low-level physical dynamics ($s \rightarrow s'$) with high-level automata state tracking, detecting high-level environmental events. (e.g., "Agent i did X").
- Rewards $r_{i,t}$ are generated by the RM transitions, making them **context/history aware**.

Nash Q-Learning

Given an agent in a one player game setting, **off policy Q-learning** method can be used to update the Q-function estimation, as a TD learning method, using the following update rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

But the max operator is **not applicable in a multi-agent setting**, where reward depends on joint behavior. This is then sobstituted by the **Nash equilibrium**, for each agent q-function:

$$Q_i(s,\mathbf{a}) \leftarrow Q_i(s,\mathbf{a}) + \alpha[r_i + \gamma Nash(Q_i(s',\mathbf{a})) - Q_i(s,\mathbf{a})]$$

Where $Nash(Q_i(s',\mathbf{a}))$ is the **expected value of agent** i when both agents follow the Nash strategy profile **at the next state** $(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}})$:

$$Nash(Q_i(s',\mathbf{a})) = \mathbb{E}_{\mathbf{a}}[Q_i(s',\mathbf{a})] = \pi_{\mathfrak{e}i}(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}})\pi_{\mathfrak{a}i}(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}})Q_i(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}})$$

Nash Q-Learning (cont'd) and Q updates

This means that, just like in standard Q-learning, the algorithm **looks ahead** to the next state $(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}})$, and **instead of looking at the max entry** with respect to all actions, it looks at **all state-action pairs and considers the average value**, weighted by **relative action choice probabilities** at the Nash equilibrium.

Process Breakdown (Lines 19–24)

$$\pi_{ji}(\cdot|s', v') \leftarrow Nash(q_{ei}(s', v'), q_{ai}(s', v'))$$
$$\overline{q}_{ji} \leftarrow \pi_{ei}\pi_{ai}q_{ji}(s', v')$$
$$q_{ji} \leftarrow (1 - \alpha_t)q_{ji} + \alpha_t(r_{j,t} + \gamma\overline{q}_{ji})$$

Additional Keypoints

Introduction of the counterfactual Reasoning

The original paper introduces the Nash Q-learning as a function of the augmented state space, i.e. $(s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$. This means that we can update the Q-function estimation for that given state using the algorithm described in the previous slide.

- Given a state $(s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$
- Given reached state $(s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}})$ though actions $a_{\mathfrak{e}}$ and $a_{\mathfrak{a}}$
- We can update as seen the **Q-function estimation for that state** $(s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$

Pointed Problem

Even in a sample game, we can reach a game state having **many different** RM internal states. Consequently, in order to perform a **general state game learning**, we're required to **experience many steps/episodes**.

Introduction of the counterfactual Reasoning (cont'd)

However, the environment-agent interaction is **independent from the agent internal RM state**: high-level events depends on game states (agent position); it's the RM that then tells the agent how it is rewarded for that event, depending on its current RM state.

So once in state s, **we don't need to experience** transitions to s' with **any possible** combination of agents RM states.

Counterfactual Reasoning

Since the **practical environment transition is independent from the RM state**, instead of updating the Q-value for state $(s, v_{\mathfrak{e}}, v_{\mathfrak{a}})$ **only**, we can

- Consider all RMs states $\tilde{v}_{\mathfrak{e}}, \tilde{v}_{\mathfrak{a}}$, simulate being in augmented state $(s, \tilde{v}_{\mathfrak{e}}, \tilde{v}_{\mathfrak{a}})$.
- Simulate state and the RM transition to $\tilde{v}'_{\mathfrak{e}}, \tilde{v}'_{\mathfrak{a}}$ if s' is reached, i.e. reaching $(s', \tilde{v}'_{\mathfrak{e}}, \tilde{v}'_{\mathfrak{a}})$
- Update the Q-value associated to $(s, \tilde{v}_{\mathfrak{e}}, \tilde{v}_{\mathfrak{a}})$

Introduction of the counterfactual Reasoning (cont'd)

Without loss of generality problems, the SGRM augmented state transition

$$(s, \tilde{v}_{\mathfrak{e}}, \tilde{v}_{\mathfrak{a}}) \to (s', v'_{\mathfrak{e}}, v'_{\mathfrak{a}})$$

we can update its correspondent Q-value estimation, together with all other augmented states in which the game state s is mapped to s', **as if** we'd experienced

$$(s, \tilde{v}_{\mathfrak{e}}, \tilde{v}_{\mathfrak{a}}) \to (s', \tilde{v}'_{\mathfrak{e}}, \tilde{v}'_{\mathfrak{a}}) \quad \forall v_{\mathfrak{e}}, v_{\mathfrak{a}} \in V_{\mathfrak{e}}, V_{\mathfrak{a}}$$

What does it means in practice?

For a $N \times M$ grid, 2 agents, 4 actions for agent, RMs with $K_{\mathfrak{e}}$, $K_{\mathfrak{a}}$ states, if standard RM based code requires E episodes to converge, then with the counterfactual reasoning we can expect to converge in $\frac{E}{K_{\mathfrak{e}} \cdot K_{\mathfrak{a}}}$ episodes.

Convergence Assumptions

Assumption 1: Exploration

Every state $s \in S$, RM states $v_{\mathfrak{e}} \in V_{\mathfrak{e}}$, $v_{\mathfrak{a}} \in V_{\mathfrak{a}}$, and actions $a_{\mathfrak{e}} \in A_{\mathfrak{e}}$, $a_{\mathfrak{a}} \in A_{\mathfrak{a}}$, are visited infinitely often when the number of episodes goes to infinity.

Assumption 2: Learning Rate

The learning rate α_t satisfies the following conditions for all t:

1. $0 < \alpha_t < 1$, $\sum_t \alpha_t(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) = \infty$, $\sum_t [\alpha_t(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}})]^2 < \infty$, and the latter two hold uniformly and with probability 1.
2. $\alpha_t(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) = 0$ if $(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}, a_{\mathfrak{e}}, a_{\mathfrak{a}}) \neq (s^t, v_{\mathfrak{e}}^t, v_{\mathfrak{a}}^t, a_{\mathfrak{e}}^t, a_{\mathfrak{a}}^t)$.

Convergence Assumptions

Assumption 3: Stage Game Structure

One of the following conditions holds during learning:

1. Every stage game $(q_{\mathfrak{e}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}), q_{\mathfrak{a}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}))$, $i \in \{\mathfrak{e}, \mathfrak{a}\}$, for all $t, s, v_{\mathfrak{e}}$, and $v_{\mathfrak{a}}$, has a **global optimal point** $(\tilde{\pi}_{\mathfrak{e}i}, \tilde{\pi}_{\mathfrak{a}i})$ (i.e., $\tilde{\pi}_{ji} q_{ji} \geq \tilde{\pi}'_{ji} q_{ji}$ for any $\tilde{\pi}'_{ji}$, $j \in \{\mathfrak{e}, \mathfrak{a}\}$) and agents' rewards in this equilibrium are used to update their Q-functions.
2. Every stage game $(q_{\mathfrak{e}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}), q_{\mathfrak{a}i}(s, v_{\mathfrak{e}}, v_{\mathfrak{a}}))$, $i \in \{\mathfrak{e}, \mathfrak{a}\}$, for all $t, s, v_{\mathfrak{e}}$, and $v_{\mathfrak{a}}$, has a **saddle point** $(\tilde{\pi}_{\mathfrak{e}i}, \tilde{\pi}_{\mathfrak{a}i})$ (i.e., $\tilde{\pi}_{ji} \tilde{\pi}_{-ji} q_{ji} \geq \tilde{\pi}'_{ji} \tilde{\pi}_{-ji} q_{ji}$ for any $\tilde{\pi}'_{ji}$, $\tilde{\pi}_{ji} \tilde{\pi}_{-ji} q_{ji} \leq \tilde{\pi}_{ji} \tilde{\pi}'_{-ji} q_{ji}$ for any $\tilde{\pi}'_{-ji}$, $j \in \{\mathfrak{e}, \mathfrak{a}\}$), and agents' rewards in this equilibrium are used to update their Q-functions.

Game Results and Experiments

Experimental Framework & Baselines

- **Testing Environment:** Evaluated on 6×6 competitive, sparse-reward grid world variations of Pac-Man. And larger 8×9 or 7×10 competitive maze maps.
- **Tasks:** Three case studies with distinct non-Markovian rewards, in the same environment:
  - `Task I`: Ego must visit its power base, than destroy Adv's power base and finally eat Adv;
  - `Task II`: Task I plus a recharge step: ego must visit its power base again once destroyed Adv's base, before eating Adv;
  - `Task III`: Task II with random origin for adv agent between two positions.
- **Comparative Baselines:**
  - `Nash-Q`: Pure spatial state-action tracking.

Case Study Empirical Analysis

- **Case Study I (Sequential Ordering):** QRM-SG converges to equilibrium by $\approx 7,500$ episodes, whereas unaugmented baselines fail entirely.
- **Case Study II & III (Energy Loops):** Explores complex recharge loops with randomized origin points. QRM-SG reaches stability within $\approx 1,000$ to $1,500$ runs.
- **Scalability:** Successfully finds the Nash equilibrium on an expanded 12×12 map space by episode 21,000.

Conclusion

Summary & Future Outlook

- **Core Contribution:** Bridges reward machines with MARL to model non-Markovian constraints in stochastic games.
- **Robust Architecture:** Augmented state transformations allow tabular actors to navigate complex environments successfully.
- **Future Directions:** Dynamically learning unknown reward machine configurations simultaneously during policy calculation; extensions to general-sum games with more agents.

References

[1] Jueming Hu, Jean-Raphaël Gaglione, Yanze Wang, Zhe Xu, Ufuk Topcu, and Yongming Liu.
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